

Systemic Lupus Erythematosus (LUPUS)

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OUTLINE

- Definition and history
- Epidemiology
- Pathophysiology
- Cases:
 - Update: Classification and diagnosis
 - Clinical Features
 - Update: Treatment



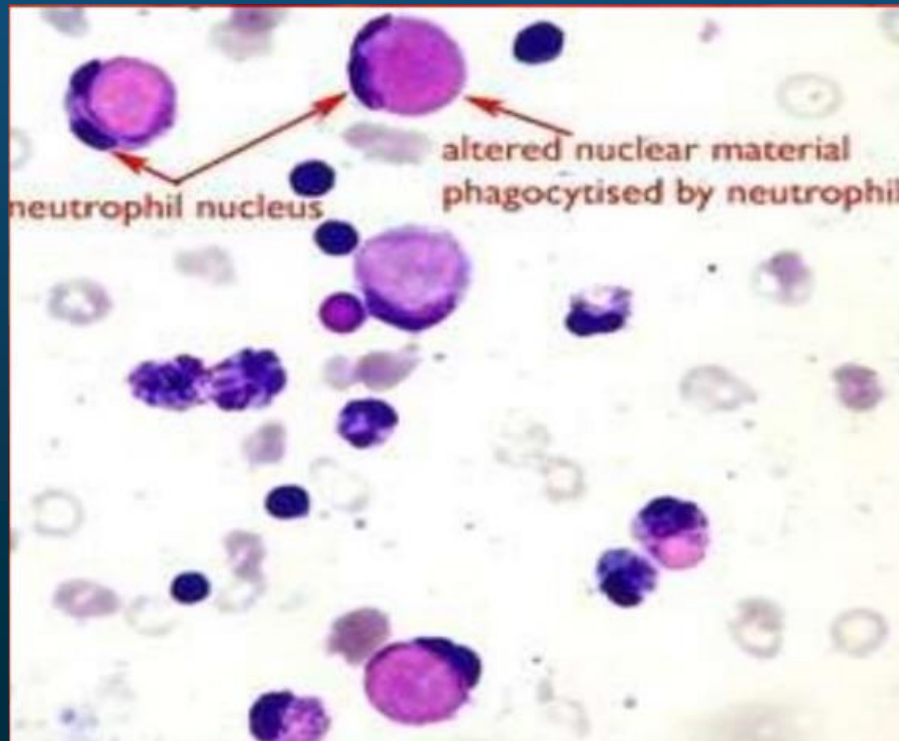
**“erythema centrifugum” and
later “lupus érythémateux”
(1838, 1850):
Laurent-Théodore Bielt
Pierre Louis Cazenave**

**“seborrhoea congestiva”
“inflammatio folliculorum”**

Figura 1 - Lupus eritematoso [tavola di Anton Elfinger, tratta da Atlas der Hautkrankheiten di Ferdinand von Hebra, Wien, 1856 (15)].

DEFINITION

- *“Inflammatory heterogeneous autoimmune disorder affecting multiple organ systems characterized by the production of auto-antibodies directed against cell nuclei”*

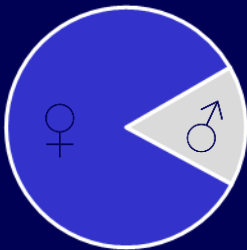


EPIDEMIOLOGY

Age, gender, race and genetics

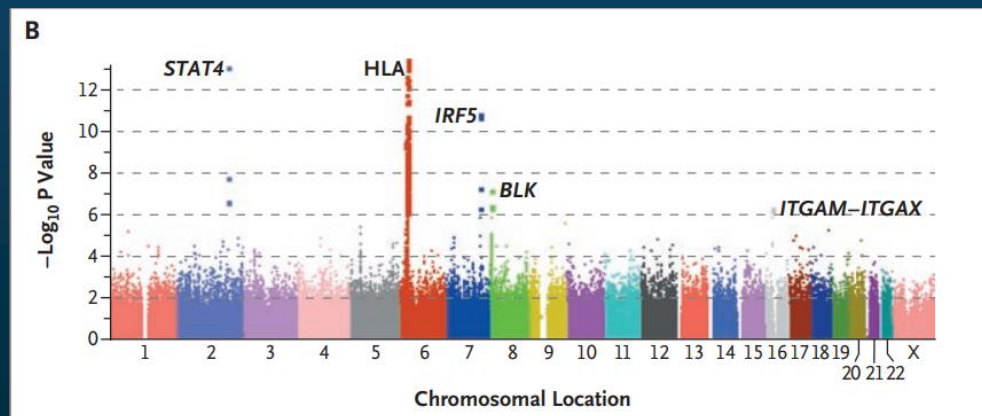
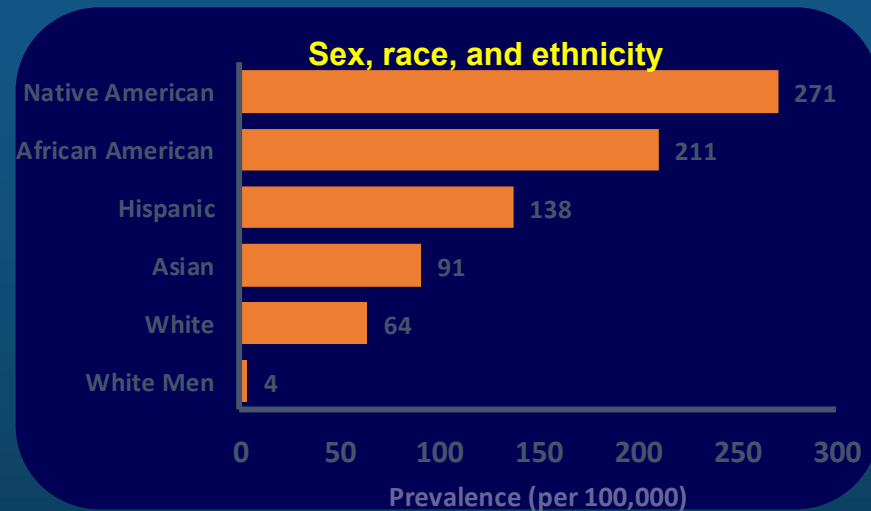
- Peak incidence 14-45 years
- Female predominance
 - severity is =
- Ethnic predisposition
- Genetics:
 - protein tyrosine phosphatase, non-receptor type 22 (*PTPN22*)
 - HLA DRB1, *ITGAM* or *ITGAX*

85% are women



Female-to-male ratio:
9 to 1

Mean age at diagnosis:
31 years

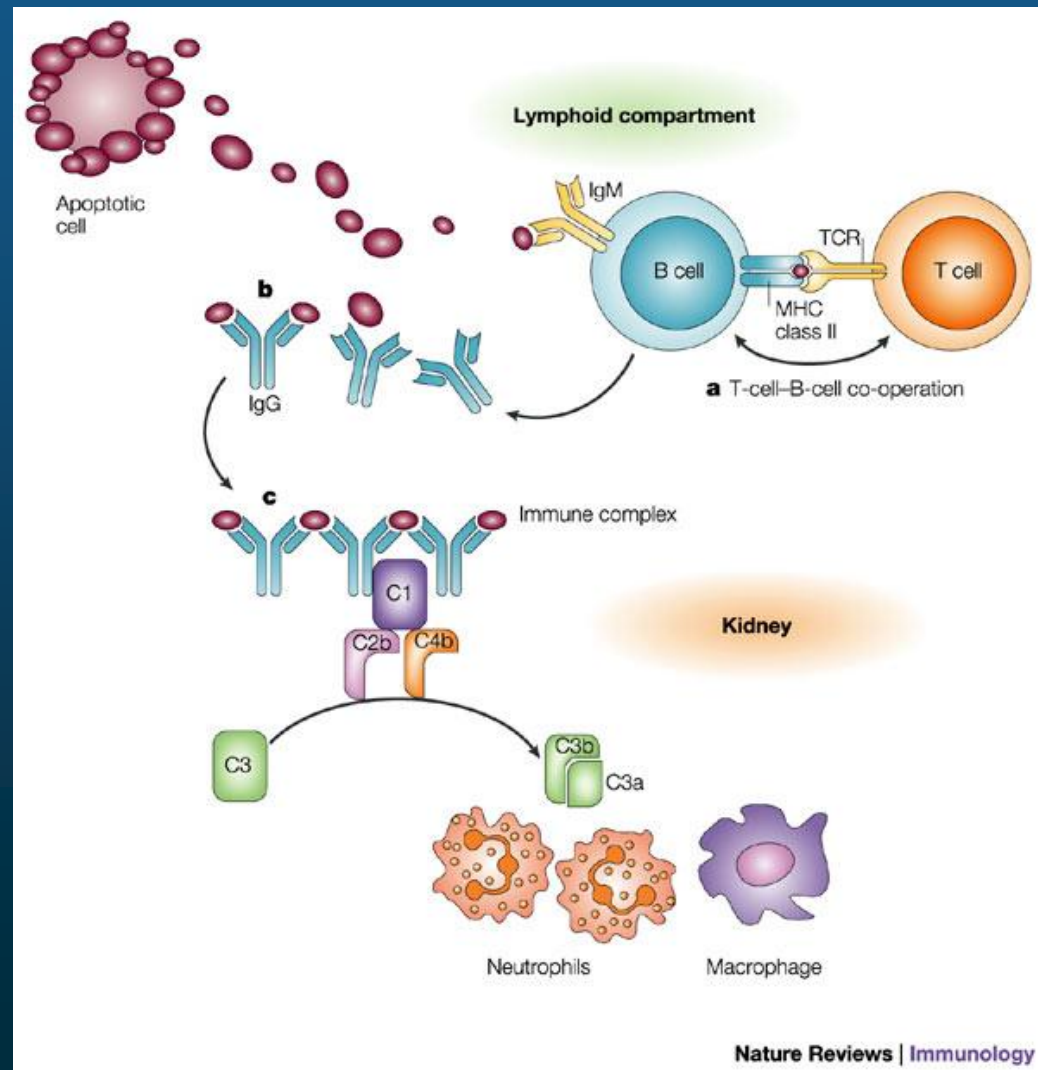


Costs of Lupus

- Lupus: 2% of all dialysis patients in the US
 - 4000 at any time
 - 1000 new/year
- US ESRD Registry 1995-2008:
 - 12,533 with lupus nephritis
 - 82% female, 49% Black
- Statistically significant findings:
 - African-American
 - Medicaid
 - underinsured
- Mortality rates
 - 50% at 7 years in 1950; now 18% at 15 years
 - Of those aged 15–24y, SLE as cause of death:
 - Ranked 10th (5th among Black and Hispanic females)

Etiology

- *Environmental*
 - *UV light*
 - *Viruses*
 - *Hormones (Estrogen)*
 - *TOBACCO*



Case 1



34 yo female presents with fevers, fatigue and a month of worsening redness and rash on cheeks and forehead; worse in sun

PMH: HTN, childhood ITP
G3P1 (2 miscarriages)

FH: mother with RA

Meds: hydralazine, amlodipine

Question 1

What lab is most specific for lupus?

anti-histone

CRP

thrombocytopenia

✓ *anti-dsDNA*

ANA

CLINICAL FEATURES: General Clinical

- *Fatigue*
- *Fevers*
- *Weight loss*
- *Anorexia*
- *Alopecia*
- *Lymphadenopathy*
- *Raynaud's*



Dermatologic domain

- Malar Rash
 - Fixed erythema; malar eminences
 - Spares the nasolabial folds

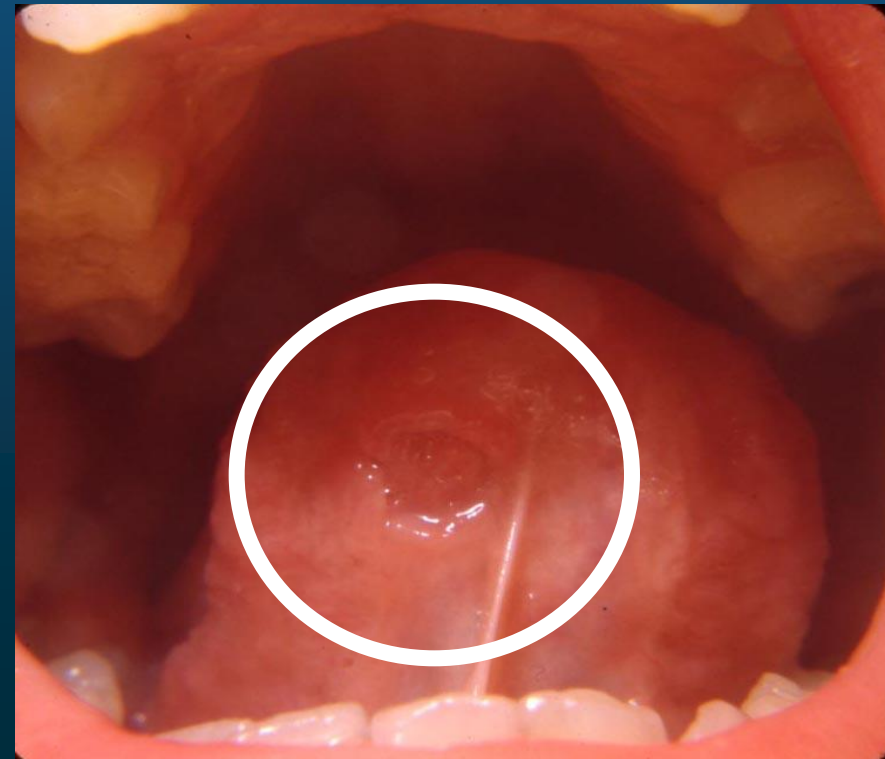
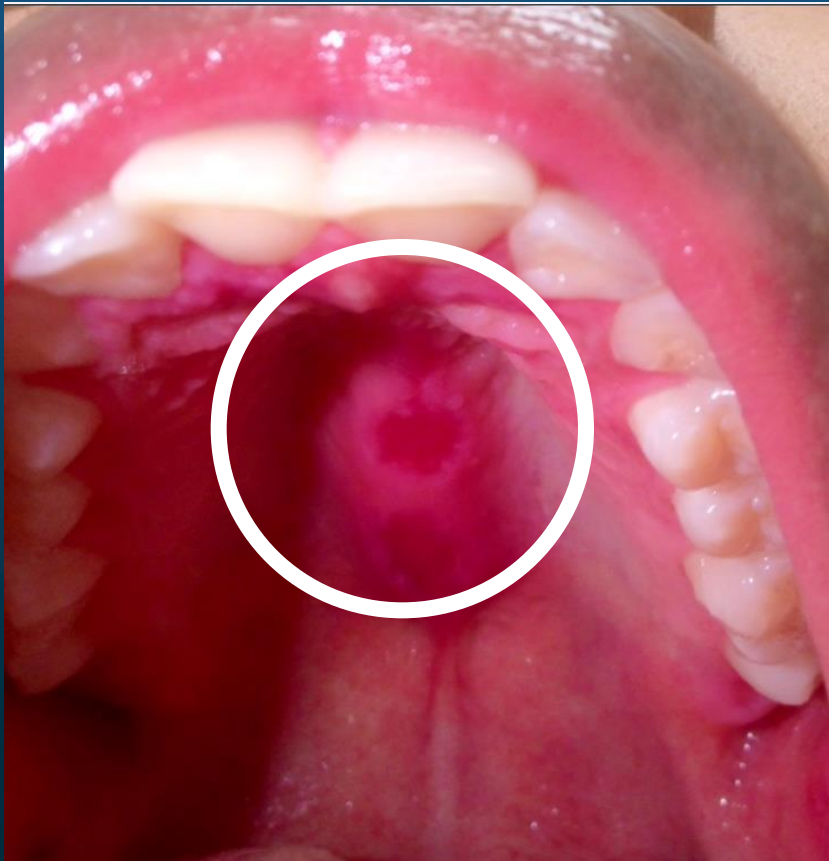


- Discoid Lupus Erythematosus (DLE)
 - Erythematous patches with central clearing
 - keratotic scaling
 - follicular plugging



ORAL ULCERS

- Oral/nasopharyngeal ulceration
- Usually painless



CLINICAL FEATURES: Musculoskeletal

- transient, small joints, symmetrical
- “Jaccoud’s” arthritis
- Most common presenting feature of SLE (90%)



Case 2

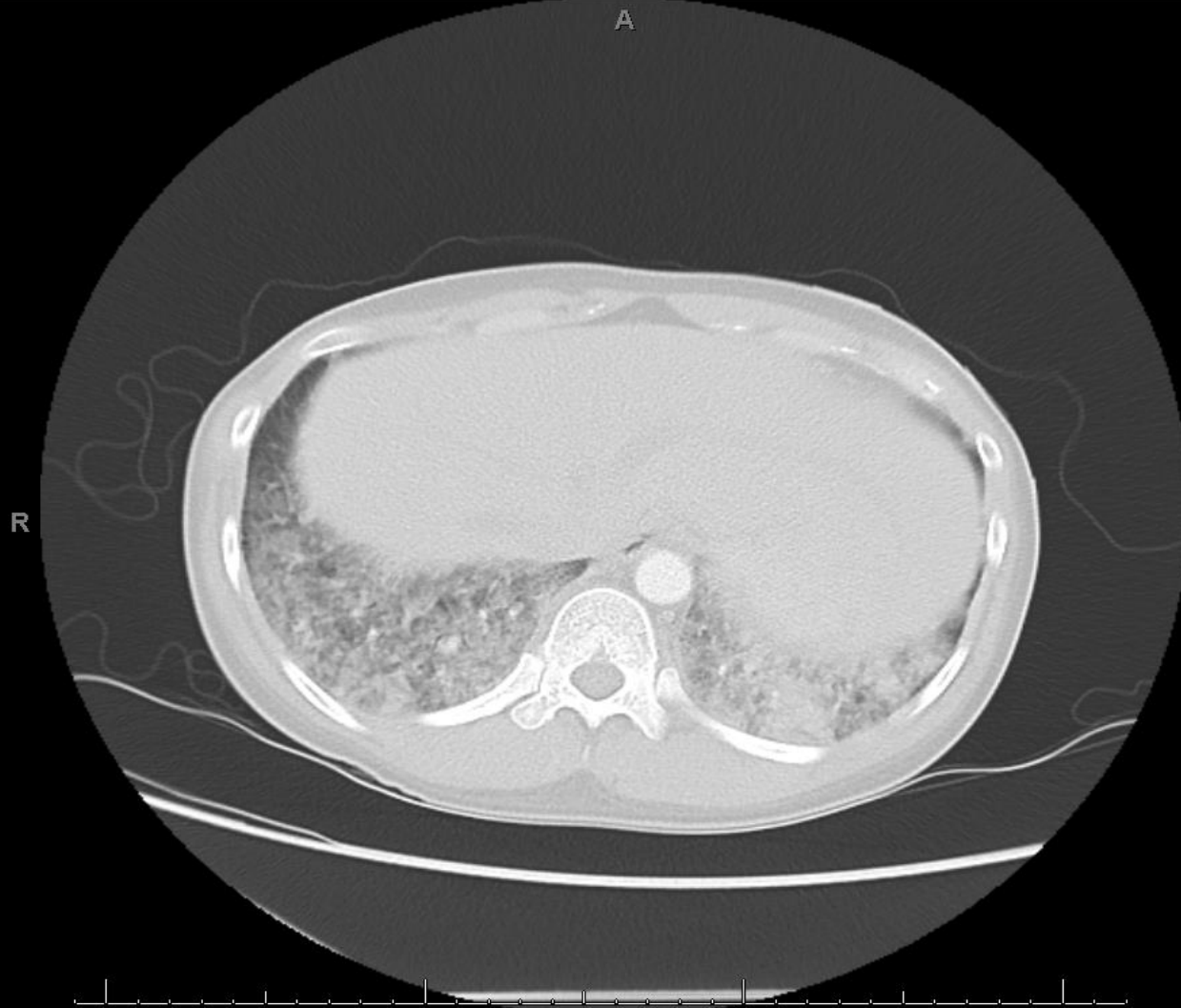
23 Cambodian female with several weeks of worsening white, painful fingers that can turn blue and red. She tried natural options including CBD, plant based diet and mindfulness as well as echinacea for “immune health”



She now presents to ED with dyspnea and hemoptysis. She is intubated. CT chest reveals....

5/15/2013 4:33:29

Kais



5/15/2013 4:33:29

Kais

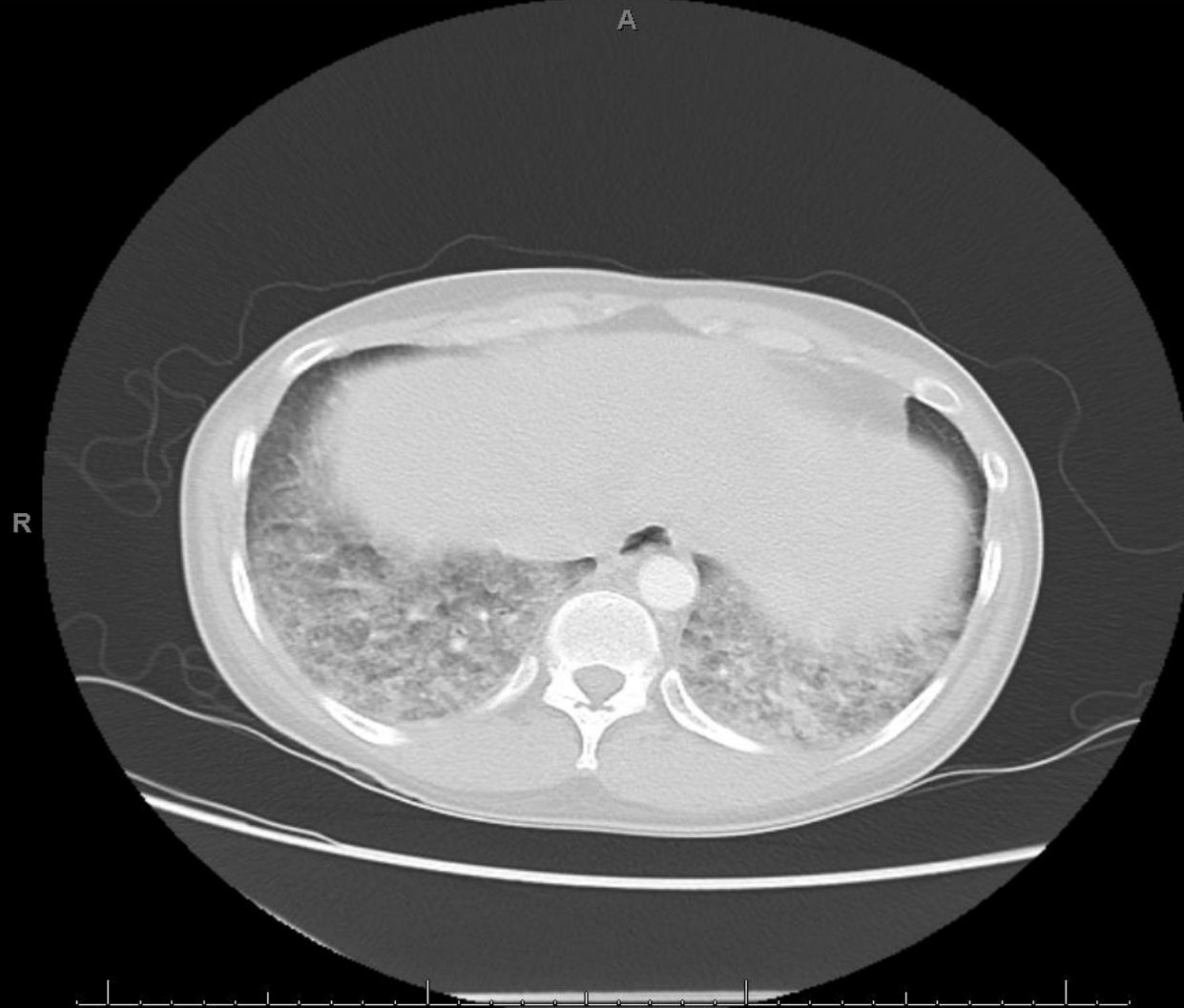
R

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5/15/2013 4:33:29

Kais

R

A

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4:48:02



Question

What is the best initial treatment for this severe presentation of lupus?

Stop echinacea

Mycophenolate

 *Methylprednisolone*

Cyclophosphamide

Plasmapheresis

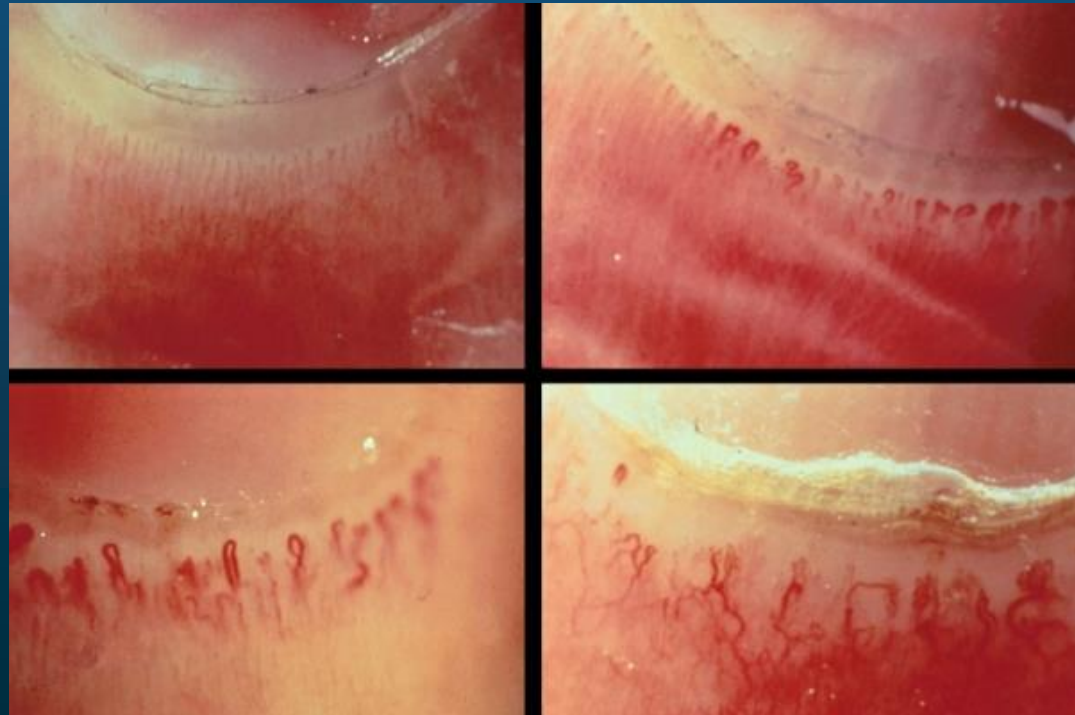
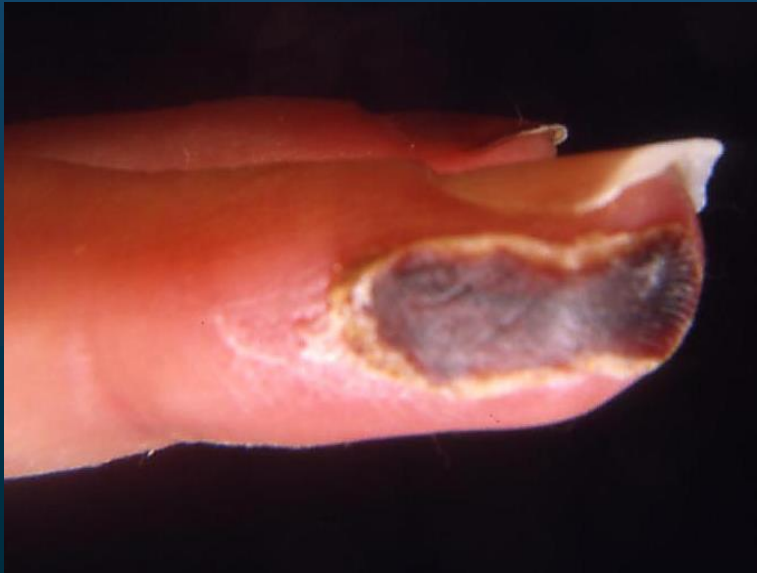
CLINICAL FEATURES: SEROSITIS

- Pleural
 - Pulmonary hemorrhage (**EMERGENCY**)
- Peritoneal
 - Mesenteric vasculitis
- Cardiac
 - Pericardial effusion

(Steroid and mycophenolate worked for my patient!)

SLE - VASCULOPATHY

- Dilated nailfold capillary loops
- Raynaud's phenomenon
- Digital ulcers



CLINICAL FEATURES: Cardiac

- Nonserosal:
 - Cardiac Arrhythmias
 - Accelerated **Atherosclerosis**
- Immunologic:
 - Libman Sacks endocarditis
 - Valvular heart disease



Noninfective thrombotic endocarditis involving mitral valve in SLE extending onto chordae tendineae.

CLINICAL FEATURES: HEMATOLOGIC DISORDER

- A) Hemolytic anemia – “AHA”
OR
- B) Leukopenia - less than $4,000/\text{mm}^3$
OR
- C) Lymphopenia - less than $1,500/\text{mm}^3$
OR
- D) Thrombocytopenia - less than $100,000/\text{mm}^3$

PANCYTOPENIA



Component <i>Latest Ref Rng</i>	4/1/2011
WBC'S AUTO 4.0 - 11.0 THOU/MCL	6.9
RBC AUTO 4.2 - 5.4 MIL/MCL	3.80 (L)
HGB 12.0 - 16.0 G/DL	12.8
HCT AUTO 37 - 47 %	36.3 (L)
PLT'S AUTO 130 - 400 THOU/MCL	235

PANCYTOPENIA



Component <i>Latest Ref Rng</i>	4/1/2011	5/13/2011
WBC'S AUTO 4.0 - 11.0 THOU/MCL	6.9	2.7 (L)
RBC AUTO 4.2 - 5.4 MIL/MCL	3.80 (L)	2.21 (L)
HGB 12.0 - 16.0 G/DL	12.8	7.8 (L)
HCT AUTO 37 - 47 %	36.3 (L)	21.2 (L)
PLT'S AUTO 130 - 400 THOU/MCL	235	10 (AA)

PANCYTOPENIA



Component <i>Latest Ref Rng</i>	4/1/2011	5/13/2011	5/13/2011
WBC'S AUTO 4.0 - 11.0 THOU/MCL	6.9	2.7 (L)	3.0 (L)
RBC AUTO 4.2 - 5.4 MIL/MCL	3.80 (L)	2.21 (L)	2.19 (L)
HGB 12.0 - 16.0 G/DL	12.8	7.8 (L)	7.6 (L)
HCT AUTO 37 - 47 %	36.3 (L)	21.2 (L)	21.3 (L)
PLT'S AUTO 130 - 400 THOU/MCL	235	10 (AA)	12 (AA)



CLINICAL FEATURES: Neurologic

Psychosis Seizures

behavior/personality changes

migraines

depression

cognitive impairment

stroke

chorea

catatonia

pseudotumor cerebri

longitudinal myelitis – neuromyelitis optica (Devic's)

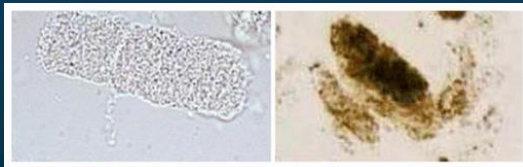
peripheral neuropathy

aseptic meningitis

*** steroid psychosis or primary psychiatric disease*

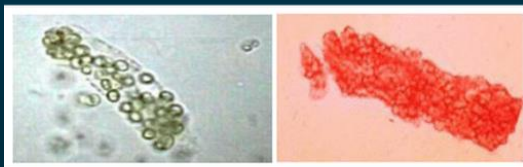
CLINICAL FEATURES: Renal (Lupus Nephritis)

- Hallmark: proteinuria (>0.5 gms daily) and **casts**
 - “Foamy” urine
 - Nephrotic syndrome
 - Hypoalbuminemia
 - Hyperlipidemia
 - Thrombophilia



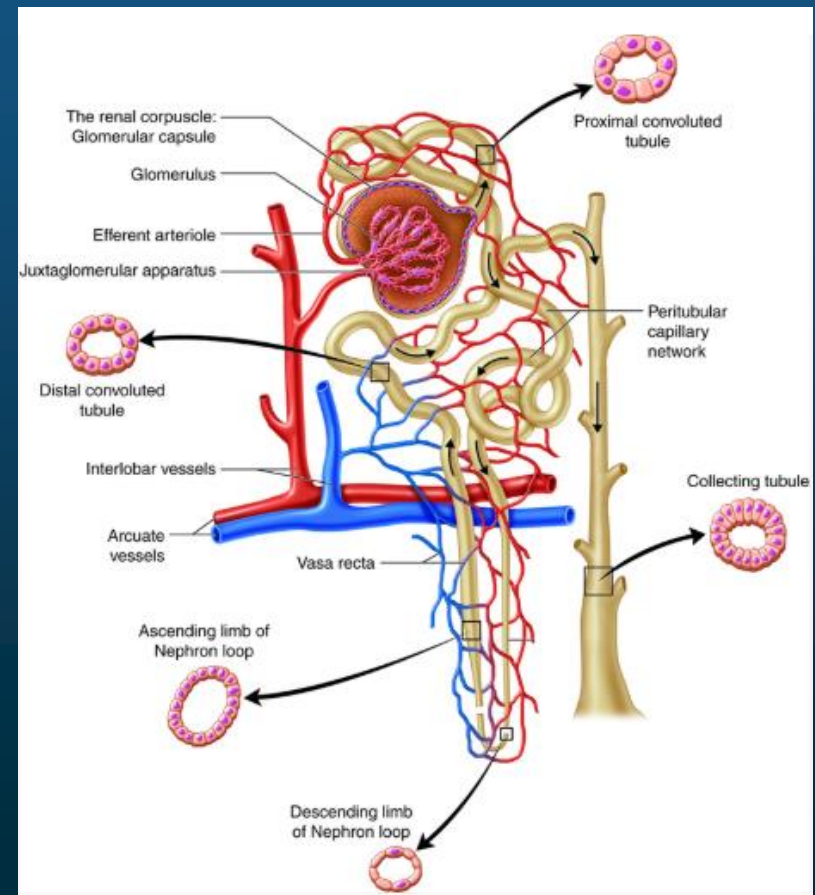
A

B



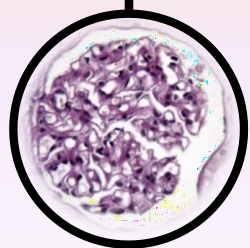
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ISN/RPS Lupus Nephritis Classification System and Prevalence (In Those Biopsied)

Class I

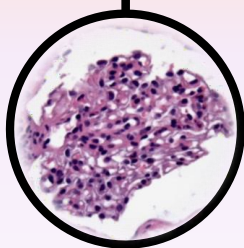


Minimal mesangial LN^{1,2}

- Normal glomeruli¹
- Mesangial immune deposits¹

Prevalence:
0.4%^{3,*}

Class II

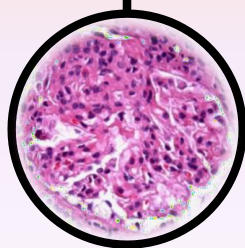


Mesangial proliferative LN^{1,2}

- Mesangial hypercellularity and/or matrix expansion¹
- Mesangial immune deposits¹

Prevalence:
11.1%^{3,*}

Class III

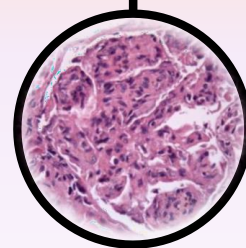


Focal proliferative LN^{1,2}

- <50% glomeruli involved, endocapillary hypercellularity^{1,4}
- Active, chronic, and/or active/chronic lesions⁵

Prevalence:
25.1%^{3,*}

Class IV

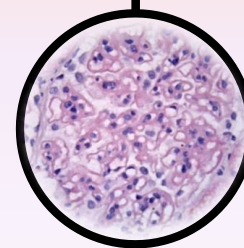


Diffuse proliferative LN^{1,2}

- IV-S – diffuse segmental LN; IV-G – diffuse global LN^{1,2}
- ≥50% glomeruli involved, subendothelial deposits¹; endocapillary hypercellularity⁴
- Active, chronic, and/or active/chronic lesions⁵

Prevalence:
53.1%^{3,*}

Class V



Membranous LN^{1,2} with global or segmental, continuous, granular, subepithelial immune deposits

- Subepithelial deposits¹
- Active or chronic glomerular lesions (Class III + V/IV + V)¹

Prevalence:
10.3%^{3,*†}

1. Kiremitci S, Ensari A. *ScientificWorldJournal*. 2014;2014:580620. doi: 10.1155/2014/580620. 2. Hahn BH, et al. *Arthritis Care Res (Hoboken)*. 2012;64(6):797-808.
3. Faezi S, et al. *Rheum Res*. 2017;2(2):51-59. 4. Bajema IM, et al. *Kidney Int*. 2018;93(4):789-796. 5. Markowitz GS, D'Agati VD. *Kidney Int*. 2007;71(6):491-495.

Immunological findings

- **ANA** - 95-100%-sensitive but highly nonspecific for SLE
- **Anti-dsDNA**-specific(60%)-specific for SLE
- 4 RNA associated antibodies
 - **Anti-Sm (Smith)**
 - Anti Ro/SSA-antibody
 - Anti La/SSB-antibody
 - Anti-RNP
- **Antiphospholipid antibodies**
 - Lupus anticoagulant-antibodies to coagulation factors. Prolonged aPTT
 - Anti-cardiolipin
 - Anti-beta 2 glycoprotein
- **Depressed serum complement (c3, c4)**
- Anti histone antibodies

2012 SLICC Classification Criteria

SLICC[†] Classification Criteria for Systemic Lupus Erythematosus

RheumTutor.com
RHEUMATISM TUTOR.COM

Requirements: ≥ 4 criteria (at least 1 clinical and 1 laboratory criteria)
OR biopsy-proven lupus nephritis with positive ANA or Anti-DNA

Clinical Criteria

1. Acute Cutaneous Lupus*
2. Chronic Cutaneous Lupus*
3. Oral or nasal ulcers *
4. Non-scarring alopecia
5. Arthritis *
6. Serositis *
7. Renal *
8. Neurologic *
9. Hemolytic anemia
10. Leukopenia *
11. Thrombocytopenia ($<100,000/\text{mm}^3$)

Immunologic Criteria

1. ANA
2. Anti-DNA
3. Anti-Sm
4. Antiphospholipid Ab *
5. Low complement (C3, C4, CH50)
6. Direct Coombs' test (do not count in the presence of hemolytic anemia)

[†]SLICC: Systemic Lupus International Collaborating Clinics

* See notes for criteria details

2019 European League Against Rheumatism/ACR Classification Criteria for Lupus

Arthritis & Rheumatology, First published: 06 August 2019, DOI: (10.1002/art.40930)

Entry criterion			
Antinuclear antibodies (ANA) at a titer of $\geq 1:80$ on HEp-2 cells or an equivalent positive test (ever)			
↓			
If absent, do not classify as SLE If present, apply additive criteria			
↓			
Additive criteria			
Do not count a criterion if there is a more likely explanation than SLE. Occurrence of a criterion on at least one occasion is sufficient. SLE classification requires at least one clinical criterion and ≥ 10 points. Criteria need not occur simultaneously. Within each domain, only the highest weighted criterion is counted toward the total score.			
Clinical domains and criteria	Weight	Immunology domains and criteria	Weight
Constitutional		Antiphospholipid antibodies	
Fever	2	Anti-cardiolipin antibodies OR	
Hematologic		Anti- $\beta 2$ GP1 antibodies OR	
Leukopenia	3	Lupus anticoagulant	2
Thrombocytopenia	4	Complement proteins	
Autoimmune hemolysis	4	Low C3 OR low C4	3
Neuropsychiatric		Low C3 AND low C4	4
Delirium	2	SLE-specific antibodies	
Psychosis	3	Anti-dsDNA antibody* OR	
Seizure	5	Anti-Smith antibody	6
Mucocutaneous			
Non-scarring alopecia	2		
Oral ulcers	2		
Subacute cutaneous OR discoid lupus	4		
Acute cutaneous lupus	6		
Serosal			
Pleural or pericardial effusion	5		
Acute pericarditis	6		
Musculoskeletal			
Joint involvement	6		
Renal			
Proteinuria $>0.5\text{g}/24\text{h}$	4		
Renal biopsy Class II or V lupus nephritis	8		
Renal biopsy Class III or IV lupus nephritis	10		
Total score:			
↓			
Classify as Systemic Lupus Erythematosus with a score of 10 or more if entry criterion fulfilled.			

CLASSIFICATION CRITERIA

- Useful for trials, but diagnosis is ultimately clinical
- Not all “Lupus” is SLE
 - Drug induced **lupus** (anti-histone antibody)
 - Anti-hypertensives (hydralazine)
 - Anti-infectives (Isoniazid, terbinafine)
 - Procainamide
 - Anti-epiletics
 - Discoid **Lupus**
 - Subacute Cutaneous **Lupus Erythematosus (SCLE)**
 - **Lupus** pernio
- Non-rheumatic:
 - HIV, HBV, HCV, endocarditis, viral infections
 - hematologic malignancies, lymphoma
 - rosacea, OA and TPO antibodies

SLE – Treatment I

- **Mild severity** (mild skin or joint involvement)
 - NSAID
 - low dose glucocorticoids
 - hydroxychloroquine
- **Intermediate severity** (serositis, cytopenia, marked skin or joint involvement):
 - glucocorticoids (1 mg/kg/day)
 - azathioprine
 - methotrexate, leflunomide
 - mycophenolate mofetil



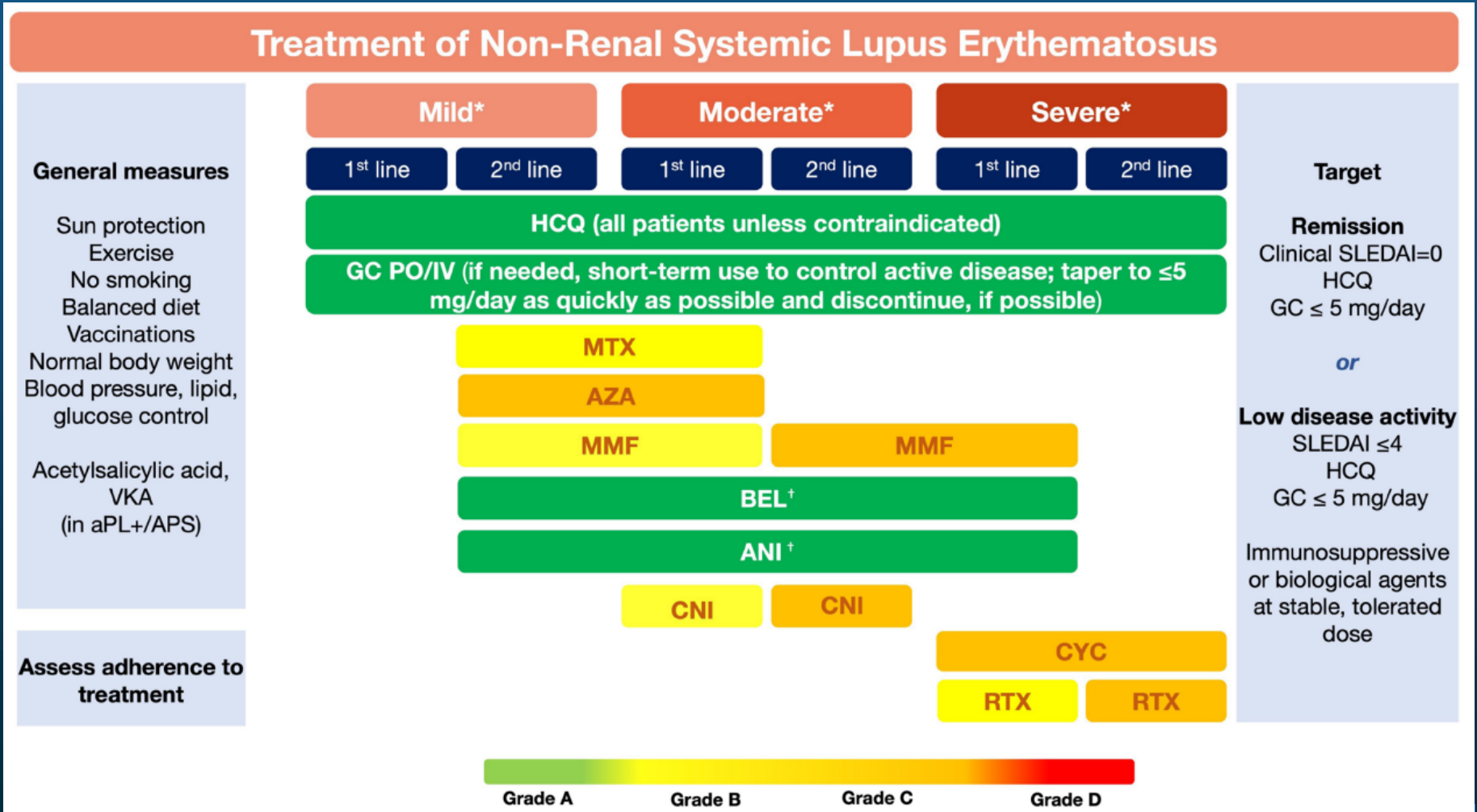
2 months of hydroxychloroquine...



SLE – Treatment II

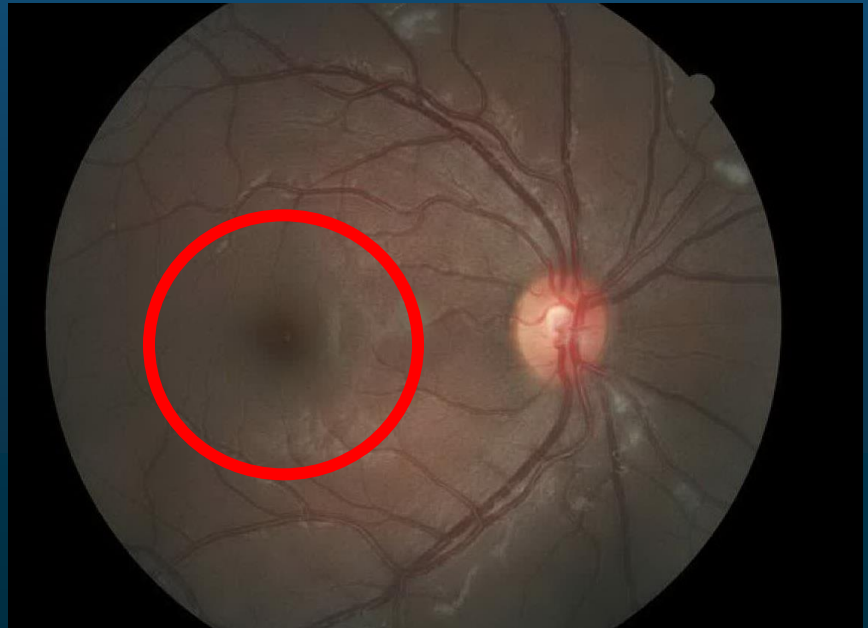
- **Severe life-threatening organ involvements** (pulmonary hemorrhage, pericarditis, nephritis, systemic vasculitis, hematologic, neuropsychiatric manifestations)
 - glucocorticoids (methylprednisolone up to 1000 mg/day x 3 days)
 - cyclophosphamide
 - plasmapheresis
 - IV immunoglobulin
 - mycophenolate mofetil
 - rituximab
 - belimumab, 2011 (FDA → lupus nephritis (LN): dec 17, 2020)
 - voclosporin (FDA → LN: jan 22, 2021)
 - anifrolumab (FDA → lupus: aug 2, 2021)
 - Obinutuzumab (FDA → LN: oct 20, 2025)

Treatment of non-renal SLE—recommended drugs with respective grading of recommendation.
aPL, antiphospholipid antibodies; AZA, azathioprine; BEL, belimumab; BILAG: British Isles Lupus Assessment Group disease activity index; CNIs, calcineurin inhibitors; CYC, cyclophosphamide; GC, glucocorticoids; HCQ, hydroxychloroquine; IM, intramuscular; MMF, mycophenolate mofetil; MTX, methotrexate; Pre, prednisone; PO, per os; RTX, rituximab; PLTs: Platelets; SLEDAI, Systemic Lupus Erythematosus Disease Activity Index.



SLE – TREATMENT PRINCIPLES

- Recent FDA approved treatments, many off label
- Recognize side effects, toxicity, infection risk and other complications
- Cholesterol, aspirin, sunscreen, ACE inhibitors, tobacco cessation, calcium, 25 OH vit D
- Teratogenicity
- Adherence
- Rare hydroxychloroquine AE:



Lupus and Pregnancy

- No increase in infertility
- Pre-conception quiescence for >6 months
- High rates of flares:
 - Preeclampsia
 - Fetal Loss
 - Preterm Delivery
 - Low Birth Weight Infant
 - DVT/PE
- Neonatal lupus and complete heart block
 - SSA



In Summary....

- Quintessential autoimmune condition
- Autoantibodies and risk factors and demographics
- Clinical domains
 - Skin and MSK
 - Renal, pulmonary, hematologic
- CBC, urine protein/creatinine, complement and anti-dsDNA
- Hydroxychloroquine, steroids, immunomodulators, biologics

THANK YOU!